

CDGCN: Conditional de novo Drug generative model using Graph Convolution Networks

Shikha Mallick¹(✉) and Sahely Bhadra²

¹ Department of Computer Science and Engineering, Indian Institute of Technology, Palakkad, Kerala, 678557, India

its.shikhamallick@gmail.com

² Department of Data Science, Indian Institute of Technology, Palakkad, Kerala, 678557, India

sahely@iitpkd.ac.in

Abstract. *De novo* drug design is a crucial part of drug discovery which is a highly expensive and slow process. Many deep learning methods have been proposed to automate and accelerate it. However, most of the current state-of-the-art methods are limited to generating novel drugs specific to proteins that already have known drugs or limited to generating molecules which lack certain desirable drug-like properties like high binding affinity or low binding energy. We introduce our graph generative model, CDGCN (Conditional *de novo* drug generative model using Graph Convolution Networks), for *de novo* drug generation for novel proteins, which takes as input a protein sequence of amino acids and generates novel molecular structures having desirable drug-like properties. CDGCN generates desirable molecules for a protein using a sequential decoding scheme by learning the distribution of generation paths of its ligands. We show that CDGCN can quickly generate novel and chemically valid drug-like molecules which have a higher binding affinity with their target proteins as compared to the state-of-the-art methods. The best binding energy between a novel protein and its novel drug-like molecules generated by CDGCN was observed to be at least -7.3 kcal/mol whereas for the state-of-the-art method it was observed to be -6.2 kcal/mol.

Availability and implementation: Code and data are available at <https://github.com/mshik/CDGCN>.

Keywords: Graph convolution network · De novo drug design · Graph generation.

1 Introduction

Drug discovery and development is an integral part of the healthcare system, with the objective to find better drugs for known diseases and effective drugs for new diseases. Usually behind a disease there is a target *protein* which is a type of biological molecule and behind a drug there is a *lead* which is a biological molecule with drug-like properties [17,39,8,10] like high binding affinity with the target protein to suppress its function [10], low synthetic accessibility score

[8], and others [17,39]. Finding optimum leads turns out to be the main part of drug discovery and is mainly done in two ways: i) Drug repurposing which is a process of finding leads from the set of known drugs which inhibit already known proteins similar to the target protein, and ii) *de novo* drug design which aims to discover novel drugs for known or novel target proteins. Drug repurposing requires detailed knowledge about the target proteins to find similar known proteins [35]. Moreover, it restricts the search space among known drugs which can hinder finding leads with better binding affinity. From now on, we mention proteins known to interact with some known drugs as known proteins, and proteins not known to interact with any known drug as novel proteins.

In this work, we focus on *de novo* drug design which is an iterative task and has multiple stages starting from i) understanding the target protein, ii) generating and optimizing leads having drug-like properties, iii) lab trial to obtain drugs from leads etc. The latter is a highly expensive and slow process and its success depends on the quality of leads found in stage ii. Efficiently finding optimum leads is not an easy task as one needs to search quite extensively among possible molecules which have drug-like properties and high binding affinity and the size of the drug-like molecule space is estimated to be 10^{60} , and the number of synthesizable molecules is in the order of 10^8 [10,33].

In recent years, machine learning methods have been introduced for *de novo* drug design, many of which utilize deep generative models [15,22,16,14,10]. The primary idea behind these methods is to generate leads with the required properties of valid drugs. The set of generated leads are significantly small to be considered for further testing which reduces the search time. These generative models learn the distribution of molecules and then generate novel molecules by sampling from the learned distribution [40]. Popular deep generative models for *de novo* drug generation can be broadly divided into two categories: i) SMILES-based methods [10] which generate SMILES strings [44] of molecules and ii) graph-based methods [15,22,16,14] which generate graph structures of molecules. There are existing methods that generate novel leads for known proteins by learning to generate molecules similar to known drugs for those proteins. The conditional graph generative model by Li et al.[22] is designed to generate novel drugs for a predetermined set of known proteins on which it is trained. The sequential decoding scheme of this model is well suited for drug molecules of different sizes as it provides the flexibility to generate graphs of different sizes unlike the methods which generate graphs by generating fixed size adjacency matrices in one step [23]. However, this model is incapable of learning the rich structural and functional information of proteins, and therefore, fails to generate novel drugs for novel proteins. The recent SMILES-based generative model by Grechishnikova[10] addresses this issue by learning to translate protein sequence of amino acids to SMILES string of drugs. To the best of our knowledge, it is the only method that is designed to generate novel drugs for novel proteins. However, SMILES-based methods are limited to learning the SMILES syntax and grammar, rather than directly learning the molecular structure, and hence they tend to generate less number of chemically valid molecules [22]. It is also shown that

Grechishnikova[10] is comparatively much slower and requires a larger memory to generate the same number of drugs as compared to Li et al.[22].

In this work, we propose a conditional graph generative model to generate novel drug structures for novel proteins which satisfy drug-like properties and strongly bind to proteins. Our proposed model CDGCN (Conditional *de novo* drug generative model using Graph Convolution Networks) learns the relationship between protein sequences, and the structure of their known drugs, and then predicts the structure of novel drugs for novel proteins exploiting that learned relationship. CDGCN utilizes a sequential decoding scheme for generating desirable molecular graphs for a given protein by learning the distribution of generation paths for ligands of that protein. Novel proteins are more widely available as sequences of amino acids than their three-dimensional structures [10] and hence it is ideal to utilize protein sequences for training the model. The three-dimensional structures of some of the proteins for testing is obtained from Protein Data Bank (PDB) [1]. We show that CDGCN generates a smaller number of prescribed leads and while reducing search space it also prescribes completely unknown or novel leads which have better binding energy than known ligands of the tested proteins, namely, i) Nitric oxide synthase, inducible (PDB code 4NOS), ii) Interleukin-1 receptor-associated kinase 4 (PDB code 2NRU), iii) Tyrosine-protein phosphatase non-receptor type 1 (PDB code 1LQF), iv) Nitric oxide synthase, brain (PDB code 1K2R), and v) Integrin alpha-L (PDB code 1CQP). The contribution of this work is as follows:

- We developed the first graph generative network, CDGCN (Conditional *de novo* drug generative model using Graph Convolution Networks), for protein-specific *de novo* drug generation for both known and novel proteins.
- As technical contribution, we propose a non-trivial way to include protein function information into the graph generation process, which allowed us to generate novel drugs having high binding affinity to novel proteins.
- Our proposed method CDGCN achieved superior performance when compared to the state-of-the-art method by Grechishnikova[10] for the task of novel protein-specific *de novo* drug generation for five novel proteins such that the best binding energy between each protein and its generated molecules by CDGCN is at least -7.3 kcal/mol whereas the same for the generated molecules by Grechishnikova[10] is -6.2 kcal/mol.

To the best of our knowledge, there is no existing method that can directly generate a reasonable number of drug-like molecules using the information from novel proteins in a short amount of time. This is crucial to reduce the time and cost required in drug discovery for novel proteins.

2 Materials and Methods

2.1 Molecular Graph Generation

A drug molecule is represented as a graph $G = (V, E)$ where V is the set of nodes in G corresponding to the atoms in the drug molecule, and E is the set

of edges in G corresponding to the bonds in the drug molecule. Each node has an atom type from a fixed set $\mathcal{A}$ of atom types and each edge has a bond type from a fixed set $\mathcal{B}$ of bond types. CDGCN takes as input a protein as a sequence of amino acids and sequentially generates a molecular graph by adding a node or an edge incrementally. The graph generation process starts with an empty graph. At each generation step, CDGCN selects one out of four different types of actions to be performed on the intermediate graph. i) *init* action which adds the first node to the empty graph, ii) *append* action which attaches a new node with a new edge to an existing node in the intermediate graph, iii) *connect* action which adds a new edge between an existing node and the last appended node in the intermediate graph, and iv) *end* action which ends the sequential graph generation process. Figure 1 shows an overview of CDGCN. In

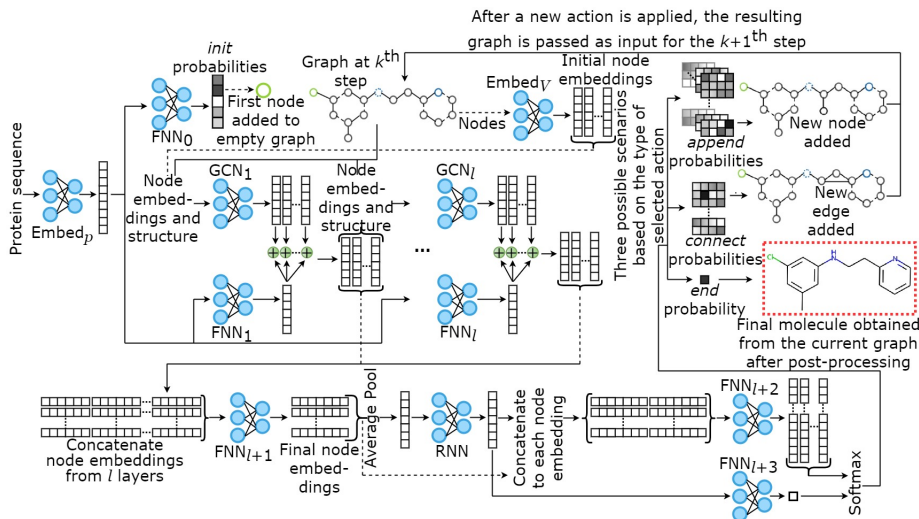


Fig. 1. Overview of our generative model CDGCN. The three types of actions, i.e., *append*, *connect* and *end*, are shown separately for better visualization.

CDGCN, $Embed_p$ takes a protein sequence and gives its embedding which contains useful information about its function. FNN_0 takes this protein embedding and gives probability scores for each atom type in $\mathcal{A}$ for the *init* action. FNN_1 , ..., FNN_l also take the protein embedding and give task-specific protein embeddings. At each generation step, $Embed_v$ takes the atom type of each node in the intermediate graph and gives its embedding. GCN_1 takes the intermediate graph structure and the node embeddings and gives new node embeddings. The task-specific protein embedding is summed with the output node embedding for each FNN and GCN layer. GCN_2 , ..., GCN_l take the intermediate graph structure and the summed node embeddings from the previous layer. The outputs

of the l different summations are concatenated and passed to FNN_{l+1} to obtain final node embeddings and then passed to an **average pooling layer** to obtain a graph-level embedding. This graph embedding is passed to RNN, which stores information from the previous intermediate graphs, and its output is concatenated with each node embedding from FNN_{l+1} . The concatenated outputs are passed to FNN_{l+2} to obtain a tensor of size $|V| \times (|\mathcal{A}| \times |\mathcal{B}| + |\mathcal{B}|)$ for all possible *append* and *connect* actions. RNN output is passed to FNN_{l+3} to obtain a scalar for **the end action**. FNN_{l+2} and FNN_{l+3} outputs are passed through Softmax layer to obtain the final probability scores for the *append*, *connect* and *end* actions. An action is then sampled to be applied on the intermediate graph. After *append* or *connect* action, the new graph is given **as input to GCN₁** for the next generation step. After *end* action, the generation process stops and the current graph is post-processed to obtain the drug molecule. The inference algorithm of CDGCN is given in the supplementary.

2.2 Model Architecture

The learnable parameters of the model are denoted as Θ . Embed_p is based on a pre-trained network from Dallago et al.[5] which was trained on protein sequences for the task of predicting **protein function from its sequence**. FNN_0 is a fully connected feed-forward neural network which consists of i) dense layer, and ii) Softmax activation [7]. **CDGCN is first pre-trained to generate chemically valid molecules without protein constraint**. In this case, a learnable weight vector followed by Softmax activation is used instead of Embed_p and FNN_0 . Embed_V is a matrix of dimension $|\mathcal{A}| \times d_0$ consisting of learnable embeddings. $\text{GCN}_1, \dots, \text{GCN}_l$ are Graph Convolution Networks (GCN) having i) Graph Convolution (GC) layer, ii) Batch Normalization (BN) layer [13,11], and iii) Rectified Linear Unit (ReLU) activation [30], except GCN_l which consists of only GC layer. GC is computed following Wu et al.[46] for molecular graphs and has the following formulation.

$$\mathbf{h}_{(v)_{\text{GC}}}^i = \mathbf{W}^i \mathbf{h}_{(v)_{\text{GC}}}^{i-1} + \sum_{b \in \mathcal{B}} \Theta_b^i \sum_{u \in \mathcal{N}_b(v)} \mathbf{h}_{(u)_{\text{GC}}}^{i-1} + \sum_{1 < d \leq D} \Theta_d^i \sum_{u \in \mathcal{N}^d(v)} \mathbf{h}_{(u)_{\text{GC}}}^{i-1} \quad (1)$$

where $\mathbf{h}_{(v)_{\text{GC}}}^i$ is the i th layer node embedding of node v . $\mathcal{N}_b(v)$ is the set of nodes which are directly connected to node v with edges having bond type b and $\mathcal{N}^d(v)$ is the set of nodes at a path length of d from the node v in the graph G . $\sum_{b \in \mathcal{B}} \Theta_b^i \sum_{u \in \mathcal{N}_b(v)} \mathbf{h}_{(u)_{\text{GC}}}^{i-1}$ and $\sum_{1 < d \leq D} \Theta_d^i \sum_{u \in \mathcal{N}^d(v)} \mathbf{h}_{(u)_{\text{GC}}}^{i-1}$, where D is the receptive field size, represent the information of the local and distant neighbourhoods of node v respectively. $\mathbf{W}^i$, Θ_b^i and Θ_d^i are learnable parameters of the i th GC layer. $\text{FNN}_1, \dots, \text{FNN}_l$ each consist of a single dense layer with output dimensions equal to those of the L GCN layers respectively. FNN_{l+1} consists of two dense layers each followed by a BN layer and ReLU activation. RNN is implemented using Gated Recurrent Unit (GRU) layers [4] as shown in the following equation.

$$\mathbf{h}_m^{\text{GRU}} = \text{GRU}_{\Theta}(\mathbf{h}_{m-1}^{\text{GRU}}, \mathbf{h}_{v^*} || \mathbf{h}_G) \quad (2)$$

where $\mathbf{h}_m^{\text{GRU}}$ is the graph-level embedding for the m th generation step containing information from the previous $m - 1$ intermediate graphs. $\mathbf{h}_{v^*}$ is the node embedding of the last appended node v^* in the current intermediate graph G . FNN_{l+2} consists of i) dense layer, ii) BN layer, iii) ReLU activation, iv) dense layer and v) exponential activation. FNN_{l+3} consisting of i) dense layer, and ii) exponential activation. $\parallel$ is concatenation.

2.3 Loss Computation and Training

The objective for CDGCN is to maximize the log-likelihood of generating a molecular graph G by following a generation path $\mathcal{T}$ given a protein P . $\mathcal{T}$ is a sequence of m actions where the j th action t_j determines the evolution of the intermediate graph G_j at the j th generation step. The next action depends on all the previous intermediate graphs and the protein P . The log-likelihood is given as

$$\log(\mathcal{P}_\Theta(G, \mathcal{T}|P)) = \sum_{j=1}^m \log(\mathcal{P}_\Theta(t_{j+1}|G_j, \dots, G_1, P)) \quad (3)$$

where m is the number of actions in $\mathcal{T}$ and G_j is the intermediate graph obtained after applying action $t_j \in \mathcal{T}$. The marginal likelihood is given as

$$\log(\mathcal{P}_\Theta(G|P)) = \log \sum_{\mathcal{T} \in S(G)} \mathcal{P}_\Theta(G, \mathcal{T}|P) \quad (4)$$

where $S(G)$ is the set of all possible generation paths. However, for most molecules encountered during drug discovery, the marginal likelihood is intractable [22]. To resolve this issue, importance sampling is used to estimate a particular probability distribution by sampling from a different predefined distribution [18,6], and hence the marginal likelihood from Equation 4 becomes

$$\log(\mathcal{P}_\Theta(G|P)) = \log \mathbb{E}_{\mathcal{T} \sim \mathcal{Q}_\alpha(\mathcal{T}|G, P)} \left[\frac{\mathcal{P}_\Theta(G, \mathcal{T}|P)}{\mathcal{Q}_\alpha(\mathcal{T}|G, P)} \right] \geq \log \frac{1}{K} \sum_{k=1}^K \frac{\mathcal{P}_\Theta(G, \mathcal{T}_k|P)}{\mathcal{Q}_\alpha(\mathcal{T}_k|G, P)} \quad (5)$$

where $\mathcal{Q}_\alpha(\mathcal{T}|G, P)$ is a predetermined distribution of generation paths in which the actions at each step are either based on canonical atom ordering [45] with probability α or based on random ordering with probability $1 - \alpha$ and K denotes the number of different generation paths possible to generate G . The detailed algorithm to compute $\mathcal{Q}_\alpha(\mathcal{T}|G, P)$ is given in the supplementary. The training of CDGCN is done in mini-batches of size N and during the training, the following negative log-likelihood loss is being minimized

$$\hat{\mathcal{L}}(\Theta) = -\frac{1}{N} \sum_{i=1}^N \log \frac{1}{K} \sum_{k=1}^K \frac{\mathcal{P}_\Theta(G_i, \mathcal{T}_{i_k}|P_i)}{\mathcal{Q}_\alpha(\mathcal{T}_{i_k}|G_i, P_i)} \quad (6)$$

3 Results and Discussion

This section reports the empirical evaluation and comparison of CDGCN with state-of-the-art benchmark methods for the quality of the generated structures for both the cases of *known* and *novel* proteins, i.e, i) the fraction of generated structures being chemically valid and novel [22], ii) the fraction of generated structures having drug-like properties [10], iii) the fraction of correctly generated structures for known proteins [22] and iv) the best binding affinity achieved by the generated structures [10]. Quantitative efficiency of CDGCN is shown by comparing the time required for generating the structures.

3.1 Datasets

Two real biological databases, i) ChEMBL [28] and ii) BindingDB [27] were used for empirical evaluation. ChEMBL is a manually curated public database of approximately 2.3 million molecules. Around 10^6 molecules that satisfy desired physicochemical properties (Section 3.4) were extracted and used for the pre-training of CDGCN and Li et al.[22]. BindingDB is a public database of experimentally measured binding energies between proteins and ligands [27]. All experiments here use the version of the BindingDB database constructed by Grechishnikova[10] where each protein is presented as FASTA sequences [26] and each ligand is presented as SMILES strings [44]. This was used for the fine-tuning of the pre-trained CDGCN and Li et al.[22], as well as for the training of Grechishnikova[10]. The BindingDB dataset is divided into five cross-validation folds where the validation folds are further split in half to obtain validation data for hyperparameter tuning and separate test data for evaluation of the trained models. The pairwise sequence similarity, measured using the Needleman-Wunsch global alignment algorithm from the EMBOSS package [36], between the proteins in the training dataset and the proteins in the test/validation dataset for each fold is kept less than 20% so that the test dataset has proteins which are truly novel for the generative model to prove its generalization. The majority of protein sequences share less than 40% similarity within the training dataset and the test dataset to make them diverse enough to train and evaluate the models. The histogram plots for the pairwise sequence similarities are given in the supplementary. Proteins in training data are known proteins whereas proteins in validation/test data are novel proteins. The models were trained, validated and tested by the ligands present in the BindingDB dataset, hence they were removed from the modified ChEMBL dataset to avoid data leakage during evaluation. In the raw dataset, each drug molecule is in its canonical SMILES format [45] which is used for obtaining the node descriptors and edge descriptors using RDKit [20]. The entire dataset, including both the modified ChEMBL and BindingDB together, has 65 unique atom types listed in the supplementary and four unique bond types (single, double, triple, and aromatic).

3.2 Baselines

For the task of generating novel drugs for known proteins, CDGCN was compared against the state-of-the-art i) graph-based method by Li et al.[22], and ii) SMILES-based method by Grechishnikova[10]. For the task of generating novel drugs for novel proteins, CDGCN has been compared with the latter.

3.3 Implementation Details

CDGCN is implemented in MXNet 1.7.0 [3] on a system with a single NVIDIA GeForce RTX 2080 Ti GPU with 8 GB memory. Training was done with batch size 8 selected using five-fold cross-validation. Batch sizes higher than 8 could not be tested due to memory constraint. It is however, worthwhile to train with higher batch sizes. All other hyperparameters and optimizer for CDGCN follow Li et al.[22]. Pre-training of CDGCN was done on the ChEMBL dataset. Finetuning of CDGCN was done on the BindingDB datasets by passing pairs of proteins and their known ligands to the model. The hyperparameters α and K from Equation 5 are selected as 0.8 and 5 respectively. A suitable choice for α and K is crucial to generate unique and novel chemically valid molecules. The hyperparameter sensitivity study is in the supplementary. Finetuning lasted on average for 10 epochs with early stopping [34] for the five different folds. Li et al.[22] and Grechishnikova[10] were trained following their original setup, however, the batch size was restricted to 8 for Li et al.[22] and the number of epochs was restricted to 6000 for Grechishnikova[10] due to memory constraint.

3.4 Evaluation Metrics

CDGCN is designed to speed up the drug discovery process, hence we evaluate its ability to quickly generate high quality molecules. We report the time required to generate proposed structures by each of the three models, i.e., CDGCN and the two baselines, along with the chemical validity of generated molecules, which is checked using RDKit [20], and novelty of generated molecules using exact string matching algorithm on the molecules in the dataset.

Along with having a good binding affinity with the target protein, a lead should also have desirable physicochemical properties [10]. Lipinski’s rule of five is a widely used rule of thumb to test whether the generated molecules have certain physicochemical properties found in orally active drugs for humans [25,32,21], like i) an octanol-water partition coefficient logP that does not exceed 5, ii) molecular weight within 500 daltons, iii) no more than 5 hydrogen bond donors (H-bond donors), and iv) no more than 10 hydrogen bond acceptors (H-bond acceptors) [24,9]. Molecules satisfying properties like i) no more than 10 rotatable bonds and ii) no more than $140\text{\AA}^2$ of topological polar surface area (TPSA) are predicted to have good oral bioavailability [41]. The quantitative estimate of drug-likeness (QED) is another property widely used to select appropriate molecules during the early stages of drug discovery [2]. It ranges from 0 to 1 with higher values indicating highly drug-like. The synthetic accessibility

score (SAS) is crucial in deciding whether a drug-like molecule is easy to synthesize [8]. We report these properties for all three models.

Following Li et al.[22], we report the percentage of "binding/active" predictions (R_c) among all generated molecules and their proteins using binary random forest classifiers. The positive class for these classifiers consisted of 100 known ligands for each known protein. The negative class consisted of 100 ligands not known to interact with that known protein, however, it might be noisy. Higher R_c is better. ECFP6 fingerprints were used as the features for the ligands [37]. We selected classifiers which had accuracy of atleast 95% for evaluation.

Leads must strongly bind to a protein to inhibit its function. We performed molecular docking to evaluate binding energies of generated molecules. We generated three-dimensional conformers for both the generated molecules and the known ligands using OpenBabel package [31]. Molecular docking is computationally heavy, therefore we selected five novel proteins with available binding sites and docked their Protein Data Bank (PDB) [1] structures with the generated molecules and their known ligands using SMINA software with default settings [19]. Jupyter Dock software [29] was used for integrating SMINA into python for parallelizing the heavy computation of the molecular docking for multiple molecules. We report binding energies for the generated molecules and the known ligands for each of the five proteins.

3.5 Experimental Evaluation

Different sets of molecules were generated by CDGCN and Li et al.[22] for each protein by repeated sampling as the generative processes here are probabilistic whereas a deterministic set of molecules was generated for each protein in the case of Grechishnikova[10]. Duplicate molecules were discarded from the sets of generated molecules. Henceforth S_N will denote the set of N generated molecules. The generation of S_N by Grechishnikova[10] was done by setting the beam size hyperparameter equal to N . To maintain consistency for evaluation with Grechishnikova[10], the molecular structures generated by both the graph-based methods were converted to canonical SMILES strings using RDKit [20]. Each experiment was repeated 10 times and the mean and standard deviations are reported. Paired t-tests for each evaluation metric were performed for statistical significance. The values in bold for multiple models indicate that the difference is not significant between those models.

Table 1 reports the i) average time in seconds required to generate the proposed structures, ii) average percentage of generated molecules that are chemically valid, novel and satisfy the constraints of all the physicochemical properties, and iii) average R_c values described in this section. Molecules generated by both the graph-based methods outperform those of Grechishnikova[10] in terms of their chemical validity, novelty and desirable drug-like properties by a large margin. This makes CDGCN computationally efficient for generating novel drugs for novel proteins which can help enlarge their sets of drugs. In the case of known proteins, Li et al.[22] is better than CDGCN for some cases but the values of CDGCN are comparable. It is important to note that more than 85%

of the molecules generated by CDGCN for both the known and the novel proteins satisfy SAS which means that CDGCN tends to generate molecules that are easy to synthesize which can further help in speeding up the drug discovery process. For the R_c values for known proteins, CDGCN is superior to Li et al.[22] but Grechishnikova[10] is superior to both of them. This is because for known proteins, Grechishnikova[10] with simple SMILES strings is easy to learn but can lead to over-fitting which reflects in Table 1 and the percentage of novel molecules generated by Grechishnikova[10] is very less as compared to the graph-based methods.

Table 2 shows the best binding energies and the percentage of generated molecules having binding energy lower than -7.0 kcal/mol for each protein. Binding energy of -6.0 kcal/mol is considered as minimum threshold for any molecule to be used for drug development [12]. In all the cases, CDGCN is superior to Grechishnikova[10]. For each of the five novel proteins, CDGCN generated molecules that gave lower binding energy than their known ligands. This shows that CDGCN is optimized to generate novel molecules which can effectively bind with a protein. Two sample molecular docking output images using PyMol [38] are provided in Figure 2.

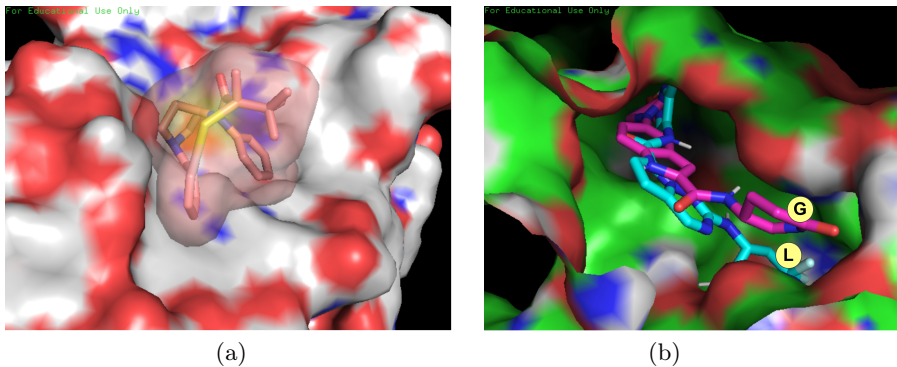


Fig. 2. (a) Docked novel protein from test dataset named Integrin alpha-L (PDB code 1CQP) having highest binding energy of -7.3 kcal/mol with its generated novel molecule by CDGCN. (b) Docked protein named Vascular endothelial growth factor receptor 2 (PDB code 1Y6A) having lowest binding energy of -11.1 kcal/mol with one of its known ligand (**L**) named ZD4190 [43] and -11.2 kcal/mol with one of its generated novel molecule (**G**) by CDGCN.

The Tanimoto similarity score, following Grechishnikova[10], shows $\sim 13\%$ of the generated molecules from all the three models-Li et al.[22], Grechishnikova[10] and CDGCN are structurally similar to the molecules in the dataset and $\sim 42\%$ of the generated molecules from all the three models differ significantly from the molecules in the dataset. We also tested the relevance of Embed_p from Figure 1 by computing binding energies of the five novel proteins from Table 2 with i)

Table 1. Qualitative and quantitative analysis of the three models-Li et al.[22], Grechishnikova[10], and CDGCN, for generating chemically valid, novel and drug-like molecules for the known and the novel proteins. Better values are in bold. ‘-’ indicates samples not generated due to memory constraint.

	Protein type	Model	Average across all proteins		
			S_{10}	S_{100}	S_{1000}
Time required to generate structures (seconds)	known	Li et al.[22]	15.5±0.77	28.8±1.42	117.0±4.63
		Grechishnikova[10]	41.5±3.30	121.4±8.57	-
		CDGCN	17.8±1.62	31.0±4.05	131.1±4.87
Valid (%)	novel	Grechishnikova[10]	42.1±2.10	190.0±12.21	-
		CDGCN	20.7±0.91	35.1±1.47	141.7±5.72
		CDGCN	20.7±0.91	35.1±1.47	141.7±5.72
Valid (%)	known	Li et al.[22]	96.3±1.55	95.2±0.57	95.4±0.21
		Grechishnikova[10]	89.0±4.34	77.3±6.44	-
		CDGCN	95.3±2.49	95.0±0.66	94.9±0.37
Novel (%)	novel	Grechishnikova[10]	85.7±4.15	63.9±2.98	-
		CDGCN	94.5±0.63	94.1±0.86	94.0±0.51
		CDGCN	94.5±0.63	94.1±0.86	94.0±0.51
< 5 log P	known	Li et al.[22]	99.7±0.22	99.7±0.10	99.7±0.03
		Grechishnikova[10]	76.8±0.20	78.0±0.13	-
		CDGCN	99.7±0.27	99.7±0.11	99.6±0.03
< 500 Molecular weight (Da)	novel	Grechishnikova[10]	82.3±0.32	84.1±0.09	-
		CDGCN	99.8±0.48	99.9±0.08	99.8±0.05
		CDGCN	99.8±0.48	99.9±0.08	99.8±0.05
< 5 Number of H-bond donors	known	Li et al.[22]	78.0±4.79	76.9±2.17	76.9±1.65
		Grechishnikova[10]	63.9±6.02	48.5±9.50	-
		CDGCN	78.8±2.65	77.5±3.69	77.1±2.92
< 10 Number of H-bond acceptors	novel	Grechishnikova[10]	55.3±9.88	31.1±4.30	-
		CDGCN	77.4±4.22	76.9±2.94	77.0±2.60
		CDGCN	77.4±4.22	76.9±2.94	77.0±2.60
< 10 Number of rotatable bonds	known	Li et al.[22]	79.0±3.99	77.9±2.16	77.9±2.19
		Grechishnikova[10]	60.4±14.02	43.6±8.33	-
		CDGCN	76.9±4.31	75.8±3.93	75.5±2.76
< 140 TPSA	novel	Grechishnikova[10]	77.3±5.10	42.1±4.62	-
		CDGCN	71.9±4.35	70.9±2.07	71.0±2.10
		CDGCN	71.9±4.35	70.9±2.07	71.0±2.10
QED	known	Li et al.[22]	91.0±2.92	88.4±1.17	88.7±0.68
		Grechishnikova[10]	66.6±9.77	50.6±7.98	-
		CDGCN	88.9±5.47	87.6±1.67	87.5±0.73
< 6 SAS	novel	Grechishnikova[10]	76.5±5.29	42.7±4.25	-
		CDGCN	85.0±4.05	84.0±1.60	84.1±1.56
		CDGCN	85.0±4.05	84.0±1.60	84.1±1.56
QED	known	Li et al.[22]	93.0±2.92	90.4±1.14	90.7±0.47
		Grechishnikova[10]	79.8±6.47	60.8±8.74	-
		CDGCN	90.5±4.36	89.3±2.04	89.2±1.89
QED	novel	Grechishnikova[10]	77.8±5.23	43.8±4.35	-
		CDGCN	89.5±2.69	88.1±1.66	87.8±1.02
		CDGCN	89.5±2.69	88.1±1.66	87.8±1.02
QED	known	Li et al.[22]	83.4±5.64	81.7±1.94	81.6±1.60
		Grechishnikova[10]	67.6±12.6	50.1±8.13	-
		CDGCN	79.5±3.39	79.2±3.33	78.7±2.08
QED	novel	Grechishnikova[10]	77.4±5.38	43.5±4.14	-
		CDGCN	78.2±2.89	74.7±1.74	74.8±1.76
		CDGCN	78.2±2.89	74.7±1.74	74.8±1.76
QED	known	Li et al.[22]	89.3±3.44	86.7±1.04	87.0±0.99
		Grechishnikova[10]	66.7±9.68	49.3±7.43	-
		CDGCN	86.5±5.12	85.4±2.05	85.1±1.56
QED	novel	Grechishnikova[10]	77.2±4.67	43.5±4.47	-
		CDGCN	83.5±3.48	82.0±2.00	82.0±1.76
		CDGCN	83.5±3.48	82.0±2.00	82.0±1.76
QED	known	Li et al.[22]	0.58±0.02	0.57±0.02	0.57±0.01
		Grechishnikova[10]	0.44±0.08	0.34±0.06	-
		CDGCN	0.57±0.01	0.57±0.02	0.57±0.02
QED	novel	Grechishnikova[10]	0.49±0.06	0.31±0.04	-
		CDGCN	0.54±0.02	0.54±0.02	0.54±0.02
		CDGCN	0.54±0.02	0.54±0.02	0.54±0.02
QED	known	Li et al.[22]	93.1±2.88	90.4±1.16	90.6±0.41
		Grechishnikova[10]	82.5±5.81	63.4±8.59	-
		CDGCN	91.1±4.64	89.8±1.09	89.6±0.79
QED	novel	Grechishnikova[10]	77.8±5.23	44.4±4.37	-
		CDGCN	89.5±2.96	88.2±1.67	87.8±0.96
		CDGCN	89.5±2.96	88.2±1.67	87.8±0.96
R_c	known	Li et al.[22]	57.4±3.30	42.1±1.02	40.1±0.92
		Grechishnikova[10]	85.2±2.28	77.9±2.19	-
		CDGCN	62.1±1.84	46.7±1.25	43.9±0.94

100 molecules generated by Li et al.[22] for the known proteins, ii) 100 molecules generated by the pretrained CDGCN without Embed_p, and iii) 100 molecules generated for each novel protein by the finetuned CDGCN with Embed_p. We found that the finetuned CDGCN with Embed_p gave the best binding energies for all five novel proteins among all the three models. Detailed ablation study is provided in the supplementary.

Table 2. Binding energies of known ligands and generated molecules for CDGCN and Grechishnikova[10] for each of the five novel proteins. ‘Best’ shows the best binding energy for the docked molecules. The PDB codes of the PDB structures used for molecular docking for each protein is also reported for reference. Better values are in bold. Underlined values are better than the values for the known ligand. ‘-’ indicates values not reported due to memory constraint.

PDB code			Best	≤ -7.0 kcal/mol (%)
4nos		Known ligands	-7.9	46.2
	S_{10}	Grechishnikova[10]	-6.2	0.0
		CDGCN	-7.3±0.24	17.0±10.00
	S_{100}	Grechishnikova[10]	-7.6	2.8
		CDGCN	-8.0±0.38	12.1±2.90
	S_{1000}	Grechishnikova[10]	-	-
2nru		Known ligands	-8.8	93.5
	S_{10}	Grechishnikova[10]	-7.9	90.0
		CDGCN	-8.3±0.37	59.3±18.60
	S_{100}	Grechishnikova[10]	-8.1	52.4
		CDGCN	-8.9±0.27	58.9±4.30
	S_{1000}	Grechishnikova[10]	-	-
1lqf		Known ligands	-11.0	99.4
	S_{10}	Grechishnikova[10]	-9.3	80.0
		CDGCN	-9.5±0.26	95.7±5.30
	S_{100}	Grechishnikova[10]	-10.1	88.0
		CDGCN	-10.2±0.12	92.8±2.20
	S_{1000}	Grechishnikova[10]	-	-
1k2r		Known ligands	-9.0	94.2
	S_{10}	Grechishnikova[10]	-7.8	91.3
		CDGCN	-8.3±0.29	63.2±4.10
	S_{100}	Grechishnikova[10]	-8.2	56.1
		CDGCN	-9.1±0.24	64.6 ±2.10
	S_{1000}	Grechishnikova[10]	-	-
1cqp		Known ligands	-9.6	98.6
	S_{10}	Grechishnikova[10]	-7.9	90.5
		CDGCN	-8.1±0.26	69.5±15.20
	S_{100}	Grechishnikova[10]	-9.2	69.3
		CDGCN	-9.8±0.32	78.0±2.50
	S_{1000}	Grechishnikova[10]	-	-
		CDGCN	<u>-10.1±0.39</u>	79.7±1.20

4 Conclusion

In this work, the first deep conditional graph generative network, CDGCN, for novel protein-specific *de novo* drug generation was introduced. Computational experiments demonstrated the superiority of CDGCN as compared to the state-of-the-art method by Grechishnikova[10] in terms of binding energies of generated molecules with the corresponding proteins, percentages of valid, novel, and diverse molecules, drug-likeness, and synthetic accessibility. In CDGCN, the encoder used to encode graphs into learned embeddings utilizes graph convolution networks adapted for molecular graphs [46]. As part of our future work, the graph encoding scheme of CDGCN will be improved upon by exploring methods that provide more informative representations of molecular graphs [42,47].

References

1. Berman, H.M., Westbrook, J., Feng, Z., Gilliland, G., Bhat, T.N., Weissig, H., Shindyalov, I.N., Bourne, P.E.: The Protein Data Bank. *Nucleic Acids Research* **28**(1), 235–242 (01 2000). <https://doi.org/10.1093/nar/28.1.235>
2. Bickerton, G.R., Paolini, G.V., Besnard, J., Muresan, S., Hopkins, A.L.: Quantifying the chemical beauty of drugs. *Nature Chemistry* **4**(2), 90–98 (Feb 2012). <https://doi.org/10.1038/nchem.1243>
3. Chen, T., Li, M., Li, Y., Lin, M., Wang, N., Wang, M., Xiao, T., Xu, B., Zhang, C., Zhang, Z.: MXNet: A Flexible and Efficient Machine Learning Library for Heterogeneous Distributed Systems. arXiv e-prints arXiv:1512.01274 (Dec 2015), <https://ui.adsabs.harvard.edu/abs/2015arXiv151201274C>
4. Chung, J., Gülçehre, Ç., Cho, K., Bengio, Y.: Empirical evaluation of gated recurrent neural networks on sequence modeling. *CoRR* **abs/1412.3555** (2014), <http://arxiv.org/abs/1412.3555>
5. Dallago, C., Schütze, K., Heinzinger, M., Olenyi, T., Littmann, M., Lu, A.X., Yang, K.K., Min, S., Yoon, S., Morton, J.T., Rost, B.: Learned embeddings from deep learning to visualize and predict protein sets. *Curr Protoc* **1**(5), e113 (May 2021). <https://doi.org/10.1002/cpz1.113>
6. van Dijk, H.K., Kloek, T.: Experiments With Some Alternatives For Simple Importance Sampling In Monte Carlo Integration. *Econometric Institute Archives* 272281, Erasmus University Rotterdam (May 1983). <https://doi.org/10.22004/ag.econ.272281>
7. Dunne, R., Campbell, N.: On the pairing of the softmax activation and cross-entropy penalty functions and the derivation of the softmax activation functions. In: *Proceedings of the 8th Australasian Conference on Neural Networks*. pp. 181–185 (1997). <https://doi.org/102.100.100/221561>
8. Ertl, P., Schuffenhauer, A.: Estimation of synthetic accessibility score of drug-like molecules based on molecular complexity and fragment contributions. *Journal of Cheminformatics* **1**(1), 8 (Jun 2009). <https://doi.org/10.1186/1758-2946-1-8>
9. Ghose, A.K., Viswanadhan, V.N., Wendoloski, J.J.: A knowledge-based approach in designing combinatorial or medicinal chemistry libraries for drug discovery. 1. a qualitative and quantitative characterization of known drug databases. *Journal of Combinatorial Chemistry* **1**(1), 55–68 (Jan 1999). <https://doi.org/10.1021/cc9800071>

10. Grechishnikova, D.: Transformer neural network for protein-specific de novo drug generation as a machine translation problem. *Scientific Reports* **11**(1), 321 (Jan 2021). <https://doi.org/10.1038/s41598-020-79682-4>
11. He, K., Zhang, X., Ren, S., Sun, J.: Identity mappings in deep residual networks. In: Leibe, B., Matas, J., Sebe, N., Welling, M. (eds.) *Computer Vision – ECCV 2016*. pp. 630–645. Springer International Publishing, Cham (2016), https://doi.org/10.1007/978-3-319-46493-0_38
12. Hiremath, S., Kumar, H.D.V., Nandan, M., Mantesh, M., Shankarappa, K.S., Venkataravanappa, V., Basha, C.R.J., Reddy, C.N.L.: In silico docking analysis revealed the potential of phytochemicals present in *phyllanthus amarus* and *andrographis paniculata*, used in ayurveda medicine in inhibiting SARS-CoV-2. *3 Biotech* **11**(2), 44 (Jan 2021). <https://doi.org/10.1007/s13205-020-02578-7>
13. Ioffe, S., Szegedy, C.: Batch normalization: Accelerating deep network training by reducing internal covariate shift. In: *Proceedings of the 32nd International Conference on International Conference on Machine Learning - Volume 37*. p. 448–456. ICML’15, JMLR.org (2015), <https://dl.acm.org/doi/10.5555/3045118.3045167>
14. Jin, W., Barzilay, D., Jaakkola, T.: Multi-objective molecule generation using interpretable substructures. In: III, H.D., Singh, A. (eds.) *Proceedings of the 37th International Conference on Machine Learning. Proceedings of Machine Learning Research*, vol. 119, pp. 4849–4859. PMLR (13–18 Jul 2020), <https://proceedings.mlr.press/v119/jin20b.html>
15. Jin, W., Barzilay, R., Jaakkola, T.: Junction tree variational autoencoder for molecular graph generation. In: Dy, J., Krause, A. (eds.) *Proceedings of the 35th International Conference on Machine Learning. Proceedings of Machine Learning Research*, vol. 80, pp. 2323–2332. PMLR (10–15 Jul 2018), <https://proceedings.mlr.press/v80/jin18a.html>
16. Jin, W., Barzilay, R., Jaakkola, T.: Hierarchical generation of molecular graphs using structural motifs. In: *Proceedings of the 37th International Conference on Machine Learning. ICML’20, JMLR.org* (2020), <https://dl.acm.org/doi/abs/10.5555/3524938.3525387>
17. Kadurin, A., Nikolenko, S., Khrabrov, K., Aliper, A., Zhavoronkov, A.: drugan: An advanced generative adversarial autoencoder model for de novo generation of new molecules with desired molecular properties in silico. *Molecular Pharmaceutics* **14**(9), 3098–3104 (2017). <https://doi.org/10.1021/acs.molpharmaceut.7b00346>, pMID: 28703000
18. Kloek, T., van Dijk, H.K.: Bayesian estimates of equation system parameters: An application of integration by monte carlo. *Econometrica* **46**(1), 1–19 (1978), <http://www.jstor.org/stable/1913641>
19. Koes, D.R., Baumgartner, M.P., Camacho, C.J.: Lessons learned in empirical scoring with smina from the csar 2011 benchmarking exercise. *Journal of Chemical Information and Modeling* **53**(8), 1893–1904 (Aug 2013). <https://doi.org/10.1021/ci300604z>
20. Landrum, G., Tosco, P., Kelley, B., sriniker, gedec, NadineSchneider, Vianello, R., Ric, Dalke, A., Cole, B., AlexanderSavelyev, Swain, M., Turk, S., N, D., Vaucher, A., Kawashima, E., Wójcikowski, M., Probst, D., guillaume godin, Cosgrove, D., Pahl, A., JP, Berenger, F., strets123, JLVarjo, O’Boyle, N., Fuller, P., Jensen, J.H., Sforna, G., DoliathGavid: rdkit/rdkit: 2020_03_1 (q1 2020) release (Mar 2020). <https://doi.org/10.5281/zenodo.3732262>, <http://www.rdkit.org>
21. Leeson, P.D., Springthorpe, B.: The influence of drug-like concepts on decision-making in medicinal chemistry. *Nature Reviews Drug Discovery* **6**(11), 881–890 (Nov 2007). <https://doi.org/10.1038/nrd2445>

22. Li, Y., Zhang, L., Liu, Z.: Multi-objective de novo drug design with conditional graph generative model. *Journal of Cheminformatics* **10**(1), 33 (Jul 2018). <https://doi.org/10.1186/s13321-018-0287-6>
23. Liao, R.: Graph Neural Networks: Graph Generation, pp. 225–250. Springer Nature Singapore, Singapore (2022). https://doi.org/10.1007/978-981-16-6054-2_11
24. Lipinski, C.A., Lombardo, F., Dominy, B.W., Feeney, P.J.: Experimental and computational approaches to estimate solubility and permeability in drug discovery and development settings. *Adv Drug Deliv Rev* **46**(1-3), 3–26 (Mar 2001). [https://doi.org/10.1016/S0169-409X\(00\)00129-0](https://doi.org/10.1016/S0169-409X(00)00129-0)
25. Lipinski, C.A.: Lead- and drug-like compounds: the rule-of-five revolution. *Drug Discov Today Technol* **1**(4), 337–341 (Dec 2004). <https://doi.org/10.1016/j.ddtec.2004.11.007>
26. Lipman, D.J., Pearson, W.R.: Rapid and sensitive protein similarity searches. *Science* **227**(4693), 1435–1441 (1985). <https://doi.org/10.1126/science.2983426>
27. Liu, T., Lin, Y., Wen, X., Jorissen, R.N., Gilson, M.K.: BindingDB: a web-accessible database of experimentally determined protein–ligand binding affinities. *Nucleic Acids Research* **35**(suppl_1), D198–D201 (12 2006). <https://doi.org/10.1093/nar/gkl999>
28. Mendez, D., Gaulton, A., Bento, A.P., Chambers, J., De Veij, M., Félix, E., Magariños, M., Mosquera, J., Mutowo, P., Nowotka, M., Gordillo-Marañón, M., Hunter, F., Junco, L., Mugumbate, G., Rodriguez-Lopez, M., Atkinson, F., Bosc, N., Radoux, C., Segura-Cabrera, A., Hersey, A., Leach, A.: ChEMBL: towards direct deposition of bioassay data. *Nucleic Acids Research* **47**(D1), D930–D940 (11 2018). <https://doi.org/10.1093/nar/gky1075>
29. Moreno, A.J.R.: Angelruizmoreno/jupyter_dock: v0.2.5 (Sep 2021). <https://doi.org/10.5281/zenodo.5514956>
30. Nair, V., Hinton, G.E.: Rectified linear units improve restricted boltzmann machines. In: *Proceedings of the 27th International Conference on International Conference on Machine Learning*. p. 807–814. ICML’10, Omnipress, Madison, WI, USA (2010), <https://dl.acm.org/doi/10.5555/3104322.3104425>
31. O’Boyle, N.M., Banck, M., James, C.A., Morley, C., Vandermeersch, T., Hutchison, G.R.: Open babel: An open chemical toolbox. *Journal of Cheminformatics* **3**(1), 33 (Oct 2011). <https://doi.org/10.1186/1758-2946-3-33>
32. Oprea, T.I., Davis, A.M., Teague, S.J., Leeson, P.D.: Is there a difference between leads and drugs? a historical perspective. *Journal of Chemical Information and Computer Sciences* **41**(5), 1308–1315 (Sep 2001). <https://doi.org/10.1021/ci010366a>
33. Polishchuk, P.G., Madzhidov, T.I., Varnek, A.: Estimation of the size of drug-like chemical space based on GDB-17 data. *Journal of Computer-Aided Molecular Design* **27**(8), 675–679 (Aug 2013). <https://doi.org/10.1007/s10822-013-9672-4>
34. Prechelt, L.: Automatic early stopping using cross validation: quantifying the criteria. *Neural Networks* **11**(4), 761–767 (1998). [https://doi.org/https://doi.org/10.1016/S0893-6080\(98\)00010-0](https://doi.org/https://doi.org/10.1016/S0893-6080(98)00010-0)
35. Pushpakom, S., Iorio, F., Eyers, P.A., Escott, K.J., Hopper, S., Wells, A., Doig, A., Williams, T., Latimer, J., McNamee, C., Norris, A., Sanseau, P., Cavalla, D., Pirmohamed, M.: Drug repurposing: progress, challenges and recommendations. *Nature Reviews Drug Discovery* **18**(1), 41–58 (Jan 2019). <https://doi.org/10.1038/nrd.2018.168>
36. Rice, P., Longden, I., Bleasby, A.: EMBOSS: the european molecular biology open software suite. *Trends Genet* **16**(6), 276–277 (Jun 2000). [https://doi.org/10.1016/S0168-9525\(00\)00204-2](https://doi.org/10.1016/S0168-9525(00)00204-2)

37. Rogers, D., Hahn, M.: Extended-connectivity fingerprints. *Journal of Chemical Information and Modeling* **50**(5), 742–754 (May 2010). <https://doi.org/10.1021/ci100050t>
38. Schrödinger, LLC: The PyMOL molecular graphics system, version 1.8 (November 2015), <https://pymol.org/2/>
39. Vamathevan, J., Clark, D., Czodrowski, P., Dunham, I., Ferran, E., Lee, G., Li, B., Madabhushi, A., Shah, P., Spitzer, M., Zhao, S.: Applications of machine learning in drug discovery and development. *Nature Reviews Drug Discovery* **18**(6), 463–477 (Jun 2019). <https://doi.org/10.1038/s41573-019-0024-5>
40. Vanhaelen, Q., Lin, Y.C., Zhavoronkov, A.: The advent of generative chemistry. *ACS Medicinal Chemistry Letters* **11**(8), 1496–1505 (2020). <https://doi.org/10.1021/acsmmedchemlett.0c00088>
41. Veber, D.F., Johnson, S.R., Cheng, H.Y., Smith, B.R., Ward, K.W., Kopple, K.D.: Molecular properties that influence the oral bioavailability of drug candidates. *Journal of Medicinal Chemistry* **45**(12), 2615–2623 (Jun 2002). <https://doi.org/10.1021/jm020017n>
42. Veličković, P., Cucurull, G., Casanova, A., Romero, A., Liò, P., Bengio, Y.: Graph attention networks (2018), <https://doi.org/10.48550/arXiv.1710.10903>
43. Wedge, S., Ogilvie, D., Dukes, M., Kendrew, J., Curwen, J., Hennequin, L., Thomas, A., Stokes, E., Curry, B., Richmond, G., Wadsworth, P.: Zd4190: an orally active inhibitor of vascular endothelial growth factor signaling with broad-spectrum antitumor efficacy. *Cancer research* **60**(4), 970–975 (February 2000), <http://cancerres.aacrjournals.org/cgi/content/full/60/4/970>
44. Weininger, D.: Smiles, a chemical language and information system. 1. introduction to methodology and encoding rules. *Journal of Chemical Information and Computer Sciences* **28**(1), 31–36 (1988). <https://doi.org/10.1021/ci00057a005>
45. Weininger, D., Weininger, A., Weininger, J.L.: Smiles. 2. algorithm for generation of unique smiles notation. *Journal of Chemical Information and Computer Sciences* **29**(2), 97–101 (May 1989). <https://doi.org/10.1021/ci00062a008>
46. Wu, Z., Ramsundar, B., Feinberg, E., Gomes, J., Geniesse, C., Pappu, A.S., Leswing, K., Pande, V.: Moleculenet: a benchmark for molecular machine learning. *Chem. Sci.* **9**, 513–530 (2018). <https://doi.org/10.1039/C7SC02664A>
47. Xu, K., Hu, W., Leskovec, J., Jegelka, S.: How powerful are graph neural networks? In: *International Conference on Learning Representations* (2019), <https://openreview.net/forum?id=ryGs6iA5Km>